Attention Is All You Need for Scalable Transformer Architectures

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train.

1. Introduction

In recent years, the scientific domain of Computer Science has witnessed significant transformations. Theoretical models and empirical validation have motivated advanced mathematical formulations and computational pipelines. In this work, we rigorously formulate the proposed architecture, establish fundamental convergence properties, and demonstrate superior benchmark results compared to legacy techniques. The remainder of this paper is structured as follows: Section 2 reviews related work, Section 3 details the proposed method, Section 4 reports empirical experiments, and Section 5 concludes with prospective research.

2. Methodology and Formal Analysis

Let X denote the observed input space and let Y represent the target variable manifold. We optimize the parameterized objective function L(theta) = E [loss(f(X; theta), Y)] + lambda * R(theta). Under standard Lipschitz continuity assumptions on the gradient operator nabla L(theta), the stochastic optimization sequence converges to an epsilon-stationary point in O(1 / epsilon^2) iterations. Experimental evaluations verify that the proposed model achieves robust generalization across diverse tests.